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RESEARCH ARTICLE

Respiration-triggered olfactory stimulation reduces obstructive sleep apnea severity: A prospective pilot study

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Summary

Obstructive sleep apnea is a prevalent sleep-disordered breathing condition characterized by repetitive reduction in breathing during sleep. The current care standard for obstructive sleep apnea is continuous positive air pressure devices, often suffering from low tolerance due to limited adherence. Capitalizing on the unique neurocircuitry of olfactory perception and its retained function during sleep, we conducted a pilot study to test transient, respiration-based olfactory stimulation as a treatment for obstructive sleep apnea markers. Thirty-two patients with obstructive sleep apnea (apnea–hypopnea index ≥ 15 events per hr) underwent two polysomnography sessions, "Odour" and "Control", in random order. In "Odour" nights, patients were presented with transient respiratory-based olfactory stimulation delivered via a computer-controlled commercial olfactometer (Scentific). The olfactometer, equipped with a wireless monitoring, analysed respiratory patterns and presented odour upon detection of respiratory events. No odours were presented in "Control" nights. Following exclusions, 17 patients entered the analysis (four women, 47.4 (10.5) years, body mass index: 29.4 (6.3) kg m^{-2}). We observed that olfactory stimulation during sleep reduced the apnea–hypopnea index ("Odour": 17.2 (20.9), "Control": 28.2 (18.6), $z = -3.337$, $p = 0.000846$, BF_{10} [Bayesian Factor 10]$= 57.9$), reflecting an average decrease of 31.3% in the number of events. Relatedly, stimulation reduced the oxygen desaturation index by 26.9% ("Odour": 12.5 (15.8), "Control": 25.7 (25.9), $z = -3.337$, $p = 0.000846$, $BF_{10} = 9.522$). This effect was not linked to the severity of baseline obstructive sleep apnea markers ($\rho = -0.042$, $p = 0.87$). Olfactory stimulation did not arouse from sleep or affect sleep structure, measured as time per sleep stage ($F_{1,16} = 0.088$, $p = 0.77$). In conclusion, olfactory stimulation during sleep was effective in reducing the severity of obstructive sleep apnea markers without inducing arousals, and may provide a novel treatment for obstructive sleep apnea, prompting continued research.

Anat Arzi and Yaron Dagon contributed equally.

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KEYWORDS
apnea prediction, breathing disorders, obstructive sleep apnea, odour, olfaction, personalized medicine, sleep

1 | INTRODUCTION

Obstructive sleep apnea (OSA), a sleep-disordered breathing condition, is characterized by repetitive reductions in breathing while asleep due to collapse of the soft tissue of the upper airways (Iber, 2007). OSA is an alarmingly prevalent disorder, with conservative estimates suggesting at least 7%–9% of the general adult population are afflicted with OSA to some degree (Punjabi, 2008; Senaratna et al., 2017). The impact of OSA on health and quality of life is widespread. OSA is linked with deleterious effects on daytime cognitive functions (e.g. attention [Salorio et al., 2002], memory [Wallace & Bucks, 2013] and depression [Schröder & O'Hara, 2005]), as well as increased morbidity and mortality as a result of accidents (Terán-Santos et al., 1999) and long-term cardiovascular comorbidities (Eckert et al., 2009; Young et al., 2002).

Continuous positive airway pressure (CPAP) treatment has long served as the primary treatment modality for OSA. Although effective, adherence to CPAP treatment regimen is often suboptimal. For clinical efficiency, minimally acceptable levels of adherence to CPAP require at least 4 hr of treatment per day (Weaver et al., 2007). Estimates suggest only 30%–50% of patients reach this criterion, and are often compelled to seek other treatment or go untreated (Rotenberg et al., 2016). Furthermore, chronic CPAP use is associated with side-effects, such as nocturnal awakenings, oral dryness and naso-oral soreness (Broström et al., 2010; Hoffstein et al., 1992). While some alternatives to CPAP treatment exist (Verbraecken et al., 2022), such as oral appliances (e.g. mandibular advancement devices; Holty & Guilleminault, 2010; Zaghi et al., 2016) or more invasive procedures (Certal et al., 2015), most are either less effective, come with various risks or impose a financial burden. As a result, CPAP remains the standard of care for efficient reduction of OSA markers.

In the search for an effective, minimally encumbering, non-invasive treatment for OSA, a central sensory pathway has been largely overlooked – olfactory stimulation during sleep. Olfactory stimulation offers a unique combination of traits that are particularly attractive for treating OSA. First, purely olfactory and mildly trigeminal odorants (trigeminality is defined as the ability to stimulate the branch of the trigeminal nerve nested within the olfactory epithelium; Frasnelli et al., 2006) do not awaken individuals from sleep (Arzi et al., 2010; Badia et al., 1990; Carskadon & Herz, 2004; Perl et al., 2017; Stuck et al., 2007). Furthermore, evidence suggests that olfactory stimulation during sleep has beneficial effects on sleep (Fismer & Pilkington, 2012; Gaeta & Wilson, 2022), both in healthy individuals (Kawai et al., 2018; Perl et al., 2016; Sano et al., 1998) and in clinical settings (Hardy et al., 1995; Lytle et al., 2014). Second, and critically, despite not inducing arousal, incoming olfactory sensory information is processed by the sleeping brain influencing physiology (Badia et al., 1990), cognition (Barnes & Wilson, 2014; Perl et al., 2017; Shanahan & Gottfried, 2014; Shanahan & Gottfried, 2017) and most importantly in the case of OSA – respiration (Arzi et al., 2010; Arzi et al., 2012).

One core feature of olfactory processing is the "sniff response", a sensorimotor loop in which presentation of odorants affects nasal airflow. This link between sensory input and respiration mode, established in wake humans (Johnson et al., 2003; Mainland & Sobel, 2006), is maintained during sleep (Arzi et al., 2010; Arzi et al., 2012; Arzi et al., 2014). Specifically, olfactory stimulation modulates both sniff volume and duration in subsequent respiratory cycles during sleep. In previous experimentation, we were able to modulate respiratory patterns in healthy adult participants using odours during sleep (Arzi et al., 2010). Our findings suggested that transient olfactory stimuli modulated the inhalation/exhalation ratio during sleep. Thus, given that odours covertly manipulate respiration during sleep, it is conceivable to use olfactory stimulation to mitigate breathing disorders. A few studies have explored the link between olfaction and respiration in clinical settings, such as on apnea of prematurity in preterm neonates (Edraki et al., 2013; Lim et al., 2021; Shanahan & Gottfried, 2017), but not as a treatment for adult OSA. In addition, in most studies, odour delivery relied on transient in-room diffusers or a single acute exposure before sleep, methods that are likely to become less effective due to sensory habituation (Pellegrino et al., 2017).

With this in mind, we designed a pilot study to test the efficacy of transient, respiration-based olfactory stimulation in reducing markers of OSA. Given that multiple experimental factors were tested in this design for the first time on patients with OSA (e.g. olfactometer setup – odours and duration, odour delivery decision algorithm), this study was intentionally small-scale and was therefore designated a pilot study. Specifically, we presented olfactory stimulation to patients with OSA using a computer-controlled olfactometry device customized to fit individual needs. The olfactometer monitored patients' respiratory patterns and presented the odours in a targeted manner upon detection of respiratory events.

2 | METHODS

2.1 | Patients

All procedures described in the study were approved by the ethics committees of Assuta Medical Center and Sha'are Zedek Medical Center. Thirty-two patients (7 women, 25 men, age 49.45 (11.18) years [median (IQR)]) with a preexisting diagnosis of OSA with a severity threshold defined by an apnea–hypopnea index (AHI) ≥ 15 events per hr of polysomnography (PSG) were recruited, and provided written informed consent to participate in the study.

TABLE 1 Inclusion and exclusion criteria for participation in the study.

Inclusion criteria	Exclusion criteria
1. OSA diagnosed; AHI ≥ 15 2. Male and female aged 30–70 years old 3. Normal olfaction; self-reported normal sense of smell 4. Patients of childbearing potential must agree to use methods of contraception and have a negative pregnancy test at screening Effective methods of contraception must be used throughout the study 5. BMI < 35 kg m^{-2} 6. Patient is willing and able to give his/her written informed consent	1. Chronic lung disease (including COPD, asthma) 2. Congestive heart failure 3. Exhibiting any flu-like or upper respiratory illness symptoms at the time of assessment 4. History of severe nasal allergies or sinusitis or difficulty breathing through the nose 5. Persistent blockage of one or both nostrils 6. Any previous operation or trauma to the nose from the last 5 years 7. Previous diagnosis of insomnia, narcolepsy, periodic limb movement disorder, respiratory failure 8. Any use of antipsychotic, hypnotic drugs 9. Major neurological diagnosis 10. Active malignant disease including chemotherapy or radiotherapy treatment 11. Pregnant or lactating women 12. Drug abuse 13. Medical history of epilepsy 14. CPAP dependency; according to investigator discretion – very high dependence on CPAP for patient's wellbeing 15. COVID-19 diagnosed from the last 3 months 16. Any known allergy to odorant capsules solution used during the study

Abbreviations: AHI, apnea–hypopnea index; BMI, body mass index; COPD, chronic obstructive pulmonary disease; CPAP, continuous positive airway pressure; OSA, obstructive sleep apnea.

Inclusion and exclusion criteria are provided in Table 1. Following exclusions, the study cohort consisted of 17 patients (4 women, 13 men, age 47.4 (10.5) years, range: 30–66 years). A CONSORT diagram summary of patient enrolment is available in Supporting Information. The demographics and clinical data of the included 17 patients are detailed in Table 2.

2.2 | Study design

This prospective, within-subject study was conducted at Assuta Medical Center and Sha'are Zedek Medical Center between October 2020 and September 2021. The study was registered online in clinicaltrials.gov (NCT04609618) and was approved by both medical centres' ethics committees (E.C. #0045-20, E.C. #0560-20).

TABLE 2 Baseline characteristics of the patient cohort.

	Mean ± SD	Median (IQR)	Range
Age (years)	49.2 ± 10.2	47.4 (10.54)	[30.3–66.5]
Weight (kg)	81.7 ± 7.9	83 (12)	[64–91]
Height (cm)	170.3 ± 8.1	172 (5)	[150–187]
BMI (kg m^{-2})	28.4 ± 3.4	29.4 (6.3)	[22.3–33.6]
AHI at referral	22.4 ± 6.7	26 (8.8)	[15–40]

Abbreviations: AHI, apnea–hypopnea index; BMI, body mass index; IQR, interquartile range; SD, standard deviation.

The study comprised of two nights: a treatment ("Odour") in which olfactory stimulation was delivered; and control ("Control"), during which participants slept in the same setting but without any odour delivery, or other OSA-related aid. The two conditions were presented in random order. Patients who were using CPAP underwent a 3-day washout period before arriving at the sleep lab for each of the experimental sessions.

2.3 | OSA severity classification

The number of apneas and hypopneas per hour of sleep were collectively termed AHI. An apnea was defined as a reduction of >90% in airflow, and a hypopnea was defined as a reduction of 30%–90% in airflow and/or effort, lasting >10 s and accompanied by an arousal or a fall in SaO_2 of ≥ 3%.

An AHI ≥ 5 per hour is a cutoff for OSA diagnosis. The AHI is further used to categorize OSA severity. Patients with AHI of 5–15, 16–30 or more than 30 events per hour are considered to have mild, moderate or severe OSA, respectively (White et al., 2015). An additional marker used in OSA evaluation is the oxygen desaturation index (ODI), which is defined as a decrease of 3% or more in blood oxygen saturation (SpO_2) below baseline. Notably, OSA severity and classification may change over time and in different sleeping conditions (Aarab et al., 2009; Fietze et al., 2004; White et al., 2015). The time gap between OSA diagnosis and participation in the experiment was 337 (396) days. Thus, to reevaluate participants' current OSA severity, we used the PSG measures recorded during the "Control" condition as a baseline for comparison with the "Odour" condition.

2.4 | Polysomnography

The PSG was conducted and scored using the SOMNOscreen PSG platform (SOMNOmedics, Germany). The same recurring staff in each centre processed the data in accordance with the guidelines of the American Academy of Sleep Medicine (AASM; Iber, 2007). PSG staff were blind to the randomization of experimental conditions. PSG montage consisted of a six-lead electroencephalogram, two bilateral electrooculogram leads, one submental lead, two tibial electromyograms and one electrocardiogram lead. Respiratory measurements in

the PSG setup were evaluated from chest wall and abdominal movement using plethysmography, nasal pressure sensors and thermal airflow signals. Oxygen saturation and heart rate were monitored using a pulse oximeter. In addition, end-tidal carbon dioxide signals were obtained ($EtCO_2$; Respironics, USA) but not used during analysis due to poor signal quality. Lastly, sleep position and audiovisual recordings were also collected.

Processed data exported from the PSG platform included sleep continuity parameters, such as latency to stage 1 and stage 2, percentage of time in wake after sleep onset (%WASO), index of awakenings, total sleep time (TST), and percentage of time in bed (%TIB). Sleep architecture parameters consisted of time spent in stage N1, stage N2, rapid eye movements (REM) and slow-wave sleep (SWS), index of sleep stage changes, and REM-onset latency. Arousals were first labelled automatically by the PSG sleep scoring algorithm, then verified and corrected if needed by the sleep lab personnel.

2.5 | Olfactory stimulation

2.5.1 | Odorants

All patients were presented with a fixed repertoire of five odour mixtures based on essential oils: orange, eucalyptus, lavender, frankincense-cinnamon and geranium (Phyto-Active, Israel). All odour materials held certificates of conformity with International Fragrance Association (IFRA) standards. Across the cohort, only a single case of a mild adverse event (nausea) was recorded, and has been adjudicated as possibly related to the olfactory stimulation.

2.5.2 | Olfactometer

Olfactory stimulation was carried out by using a computerized olfactometer – commercial product designed to present odours at predefined times (Scentific™, AppScent Medical). The olfactometer consisted of the following main components: a respiration detector, a controller, an odour disperser and an odour cartridge (Figure 1a). The respiration detector was equipped with a diagnostic system capable of online monitoring of patients' respiratory patterns based on Impulse Radio Ultra-Wideband (IRUWB) radar (Heglum et al., 2021; Kang et al., 2020; Pallesen et al., 2018). The controller unit detected movement patterns associated with the onset of apnea–hypopnea epochs that then triggered odour delivery. Finally, the odour disperser consisted of a computer-controlled compressed air compartment in which odorant cartridges were housed.

We positioned the disperser at about 50 cm from the patient's head to target odour delivery at the vicinity of the nose. This setup optimized olfactory stimulation devoid of other visual, tactile, humidity or thermal stimuli.

2.5.3 | Odour delivery regimen

The olfactometer constantly monitored respiratory patterns. Upon detection of apnea by the algorithm, an odour was presented approximately 1 s later for a period of 20 s, concluding with an interstimulus interval (ISI) of 40 s. Apnea events did not trigger odour presentation if they occurred within the ISI. The olfactometer randomly rotated between the five odours, to prevent olfactory habituation. Following each odour burst, non-odorous room air was emitted from the nozzles for 3 s to reduce odour contamination to the tubing (Figure 1b). Apnea detection was based on a decision-tree algorithm, optimized to a 20-feature repertoire of the IRUWB signal. Critically, the olfactometer did not counteract events only once they met the minimal duration of 10 s as clinically defined, but attempted to constantly predict their occurrence.

3 | DATA ANALYSIS AND STATISTICAL APPROACH

3.1 | Power analysis

A power analysis, computed on a preliminary pilot cohort ($N = 5$), suggested a decrease of 40% ± 27.8% in AHI (Mean ± SD). In light of these results, we conducted a power analysis aimed at 5% alpha and 80% statistical power, which revealed that a sample size of $N = 20$ would be a suitable cohort to demonstrate such effects. We predicted that ~40% of participants could potentially be discarded, due to failure to complete the 2-night course of the paradigm, difficulties falling asleep in the sleep clinic or AHI < 15 in the "control" night due to variability in OSA severity over time. Thus, we opted to recruit a substantially larger cohort.

3.2 | Statistical design

As a first step, the data underwent tests for normality using the Shapiro–Wilk test. We observed that the majority of parameters, more specifically the OSA-related measures, did not meet normality criterion (AHI: $SW = 0.872$, $p = 0.0016$; ODI: $SW = 0.824$, $p = 0.0005$). Therefore, we used non-parametric tests. All analyses described in this study were conducted using two-tailed tests. Analyses were carried out using custom code in MATLAB R2018a (MathWorks, Natick, MA, USA) and JASP 2019 version 0.16.4.

Additionally, we included Bayesian non-parametric paired tests where appropriate, to report the strength of the evidence in favour of the null hypothesis (Gliner et al., 2002). Specifically, we compared the probability that odour presentation had no effect on a parameter (H0) with the alternate hypothesis that this parameter will change during the "Odour" condition (H1) with a Cauchy prior of 0.707 (Jeffreys, 1998). In line with the frequentist hypothesis testing, Bayesian tests also relied on two-sided assumptions. All Bayesian statistical analyses were conducted in JASP 2019 version 0.16.4.

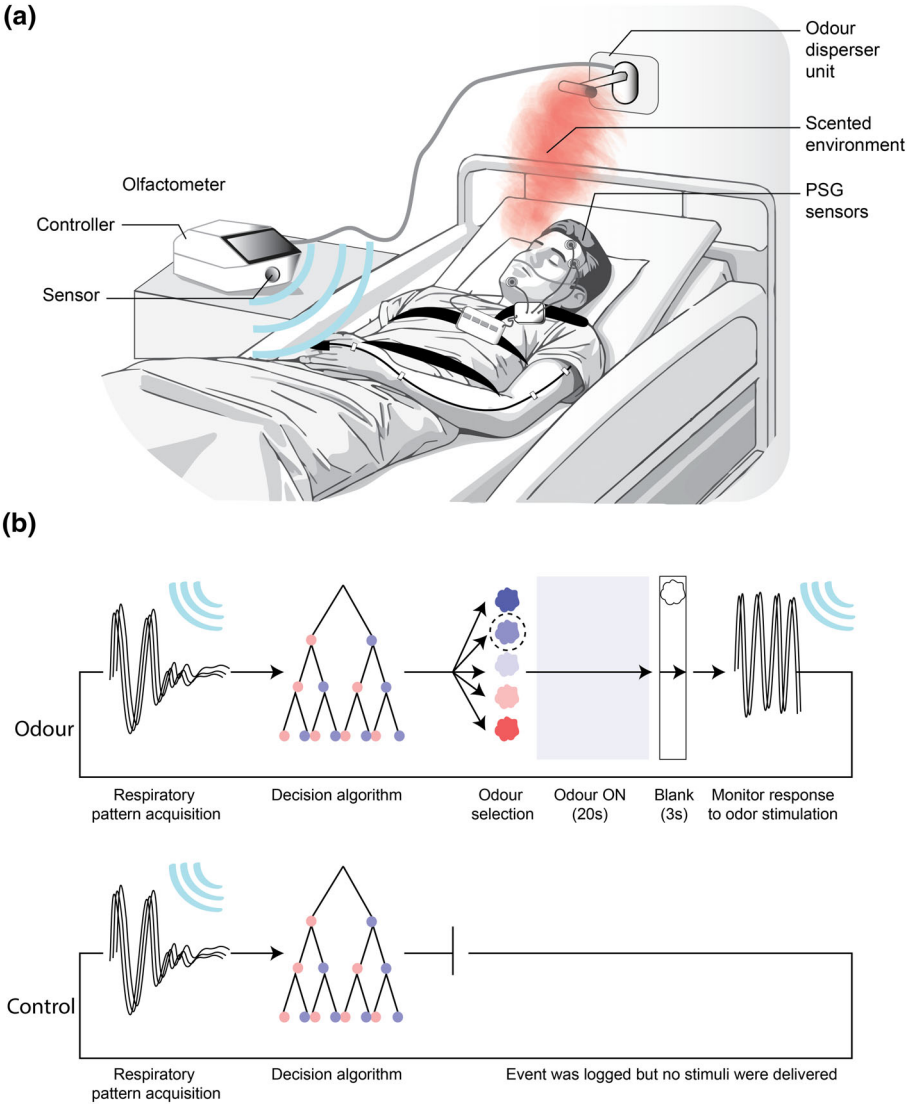

FIGURE 1 (a) Illustration of the experimental room. Patients spent two nights undergoing standard polysomnography (PSG). The olfactometer's controller unit monitored and analysed respiratory patterns using a contactless radio wave emitter. Olfactory stimulation was delivered via a wall-mounted disperser unit above the patient's head. (b) Olfactory stimulation procedure. Top: in the "Odour" condition, respiratory patterns were continuously monitored and served as input to the decision algorithm. If characteristics of apnea–hypopnea patterns were met, the olfactometer dispersed one of five odorants selected at random for 20 s. Next, a 3-s burst of clean air (Blank) was presented. Finally, the olfactometer again monitored respiratory patterns for the next bout of this decision cycle. Bottom: in the "Control" condition, respiratory patterns were continuously monitored and served as input to the decision algorithm in order to log respiratory events, but no odours were dispersed. Note that contrary to the illustration, wireless monitoring did not stop during odour/blank delivery but persisted during odour presentation.

4 | RESULTS

4.1 | Olfactory stimulation during sleep reduced apnea–hypopnea events in patients with OSA

First, we tested whether olfactory stimulation during sleep decreased OSA markers. We observed that in nights with olfactory stimulation, AHI indices were significantly reduced (AHI "Odour": 17.2 (20.9), AHI "Control": 28.2 (18.6), Wilcoxon signed-rank test (14/17 patients): $z = -3.337$, $p = 0.000846$, $BF_{10} = 57.9$), reflecting an average decrease of 31.3% in the number of events (Figure 2a). We then tested whether olfactory stimulation had sleep-stage-specific beneficial effects by calculating sleep-stage-specific AHI (the frequency of apnea–hypopnea events in each sleep stage), and contrasted these results between "Odour" and "Control" nights. We observed that without olfactory stimulation, apnea–hypopnea events occurred predominantly in N2 sleep (AHI = 33.5 (18.1)) and REM sleep (AHI = 29.1 (22.9)), and less frequently in N3 sleep (AHI = 5.3 (30.2)). When odour stimulation was applied, the stage-specific AHI was significantly reduced during N2 sleep ("Odour": 16.28 (32.4), "Control": 33.54 (18.1), Wilcoxon signed-rank: $z = -2.34$, $p = 0.019$, $BF_{10} = 2.47$) and N3 sleep ("Odour": 2.26 (4.7), "Control": 5.36 (30.23), Wilcoxon signed-rank: $z = -2.79$, $p = 0.032$, $BF_{10} = 2.08$). The same trend was observed during REM sleep, but did not reach significance ("Odour": 21.73 (26.4), "Control": 29.06 (22.9), Wilcoxon signed-rank: $z = -1.63$, $p = 0.102$, $BF_{10} = 0.80$). Finally, we compared the effects across sleep stages by directly contrasting odour-mediated difference (["Odour" – "Control"]), and observed that effects in non-REM sleep were not significantly higher than REM sleep ("N2" versus "REM" Wilcoxon signed-rank: $z = -0.83$, $p = 0.431$, $BF_{10} = 0.31$; "N3" versus "REM" Wilcoxon signed-rank: $z = -0.64$, $p = 0.548$, $BF_{10} = 0.35$). These results suggest that the effectiveness of olfactory stimulation in reducing AHI is independent of sleep stage.

Next, we confirmed that the chronological order of conditions had no effect on OSA markers. We employed the same analytical approach, this time modelling session order ("Night 1"/"Night 2") instead of conditions of olfactory stimulation. The two sessions took place 7 (3) days apart (note that time between sessions was calculated

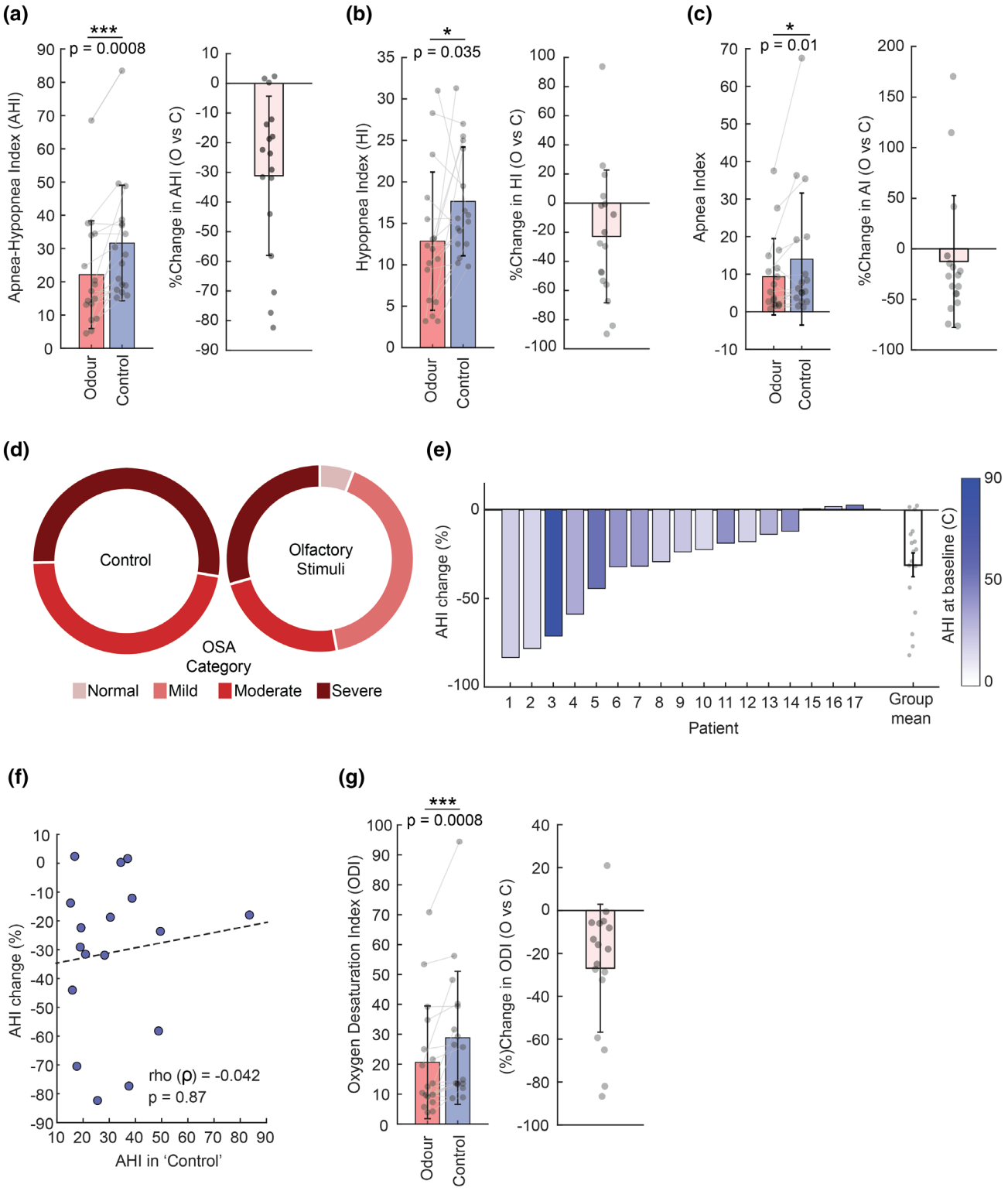

FIGURE 2 (a) Box plots depicting median-based group distribution of apnea–hypopnea index (AHI) scores for "Odour" and "Control" nights (red and blue, respectively). Points represent participants ($N = 17$), with cross-condition lines overlaid to aid in visualization of within-subject effects. Thick red bars denote group median. Dashed whiskers extend to 25th and 75th percentiles. Data points exceeding this range are marked with red crosses. $^{*}p < 0.05$, $^{***}p < 0.001$. (b) Same as (a) but for the effect of olfactory stimulation on measures of the oxygen desaturation index (ODI). (c) Same as (a) but for the effect of olfactory stimulation on the percentage of time spent in oxygen saturation (SpO_2) lower than 90%. (d) Same as (a) but for the effect of olfactory stimulation on Wake index. (e) Categorical change in apnea severity classification following olfactory stimulation during sleep ($N = 17$). A breakdown of OSA classification in the "Control" night (left) versus when "Olfactory stimuli" were presented (right). Colour intensity signifies OSA severity categories ranging from normal (light red) to severe (dark red). (f) Bars depicting individual (%) change of AHI in "Odour" nights compared with "Control" nights. Colour map denotes OSA severity according to baseline AHI measured in the "Control" night. The rightmost bar depicts (%) change of the group mean. Points represent participants ($N = 17$). Error bars are SEM.

with the exclusion of one patient who had to be summoned for a third session 177 days following their previous visit, due to a technical issue). OSA-related respiratory measures were not affected by session order (AHI: Night 1: 24.7 (22.1), Night 2: 20.4 (16.8), Wilcoxon signed-rank test (8/17 patients): $z = 0.40237$, $p = 0.68740$, $BF_{10} = 0.32$), suggesting that the reduction in AHI following odour stimulation was not a result of conditions order. Together, these findings demonstrate that transient odour stimulation during sleep decreases AHI in patients with OSA.

Following odour presentation, clinical categories of patients' OSA severity was either improved or remained stable as follows. Moderate OSA: out of the nine patients, seven improved: one (11.1%) to normal AHI (< 5 events per hr) and six (66.6%) to mild OSA, while two (22.2%) patients remained in the moderate category. Severe OSA: out of seven patients, three improved: one (12.5%) patient improved to mild OSA, and two (25%) improved to moderate OSA, while five (62.5%) patients remained in the severe OSA category (Figure 2e). Importantly, in none of the patients was OSA severity exacerbated according to clinical categories. We observed variability in the magnitude of AHI reductions following odour stimulation (Figure 2f), and therefore asked whether this variability is linked to the severity of baseline OSA markers (i.e. AHI in "Control" = 28.2 (18.6), range: 15.2–83.5). Nevertheless, these two measures were not correlated (Spearman's $\rho = -0.042$, $p = 0.87$).

In line with the improvement in AHI, we observed a reduction in ODI and oxygen saturation (SpO_2). Olfactory stimulation reduced ODI by 26.9% on average (ODI "Odour": 12.5 (15.8), ODI "Control": 25.7 (25.9), Wilcoxon signed-rank test (16/17 patients): $z = -3.3373$, $p = 0.000846$, $BF_{10} = 9.522$; Figure 2g), and reduced the proportion of time spent at SpO_2 < 90% ("Odour": 4.4 (4.9), "Control": 5.4 (14.0)%, Wilcoxon signed-rank test: $z = -2.51$, $p = 0.0121$, $BF_{10} = 3.966$; Figure 2c). In addition, a marginal reduction in the average depth of SpO_2 desaturation was observed ("Odour": 5.688 ± 1.369, "Control": 6.424 ± 2.487, Wilcoxon signed-rank test: $z = -1.94$, $p = 0.0517$, $BF_{10} = 1.586$). Mean and minimal SpO_2 were not significantly altered by olfactory stimulation (Mean SpO_2: "Odour": 93 (1), "Control": 93 (2), Wilcoxon signed-rank test: $z = -1.68$, $p = 0.125$, $BF_{10} = 0.832$; Minimal SpO_2: "Odour": 84 (4), "Control": 82 (5), Wilcoxon signed-rank test: $z = 1.13$, $p = 0.271$, $BF_{10} = 0.451$). Please refer to Table 3 for a summary of the results reported in this section. Together, these findings suggest that odour presentation during sleep reduces the severity of respiratory OSA-related markers.

4.2 | Olfactory stimulation during sleep did not alter the duration of apnea–hypopnea events in patients with OSA

Next, we tested whether olfactory stimulation during sleep reduced the duration of apnea–hypopnea events. First, we examined all five odours combined, and found no difference in event duration between "Odour" and "Control" nights ("Odour": 20 (4) s, "Control": 20 (4.5) s, Wilcoxon signed-rank test (12/17 patients): $z = -1.160$, $p = 0.263$, $BF_{10} = 0.260$). To examine the direct effect of odour stimulation, we compared the duration of apnea–hypopnea events in the "Odour" nights that were conjugated with odour stimulation (i.e. the respiratory event that was the trigger for odour delivery) with events that were not. We first assessed the olfactometer's apnea–hypopnea detection rate by comparing its events with the PSG, which underwent manual inspection by experienced sleep technicians. Because the olfactometer logged apnea–hypopnea events both in the "Odour" condition and "Control" (without applying the odour stimulation), we inspected both datasets. Overall the olfactometer presented 103 (66) odour pulses per treatment night. The olfactometer's detection rate for events marked by the PSG was 50.35 (24.26)%, and did not differ between "Odour" and "Control" nights ("Odour": 46.74 (20.04)%, "Control": 50.38 (28.61)%, Wilcoxon signed-rank: $z = -0.213$, $p = 0.854$, $BF_{10} = 0.250$). In other words, approximately half of the apnea–hypopnea events were conjugated with odour stimulation.

We found no difference in event duration between the two conditions ("Conjunction": 19.5 (5.3) s, "Non-Conjunction": 20.5 (5.18) s, Wilcoxon rank sum test (11/17 patients): $z = -0.828$, $p = 0.431$, $BF_{10} = 0.343$). Next, we examined the influence of each of the five randomly presented odour mixtures separately, by separately sampling the duration of events conjugated with each odour type. Normalized apnea–hypopnea events duration (within patient using z-score) showed no consistent effect of odorant type at the group level (rmANOVA $F_{4,64} = 0.5891$, $p = 0.672$). Given the known individual variability in odour perception (Logan, 2014; Trimmer et al., 2019), we repeated the analysis at the single-patient level. Again, we observed no differences as a function of odour type. These findings imply that odour stimulation during sleep does not reduce the duration of apnea–hypopnea events.

4.3 | Olfactory stimulation did not promote arousal or alter sleep structure

Sleep quality questionnaires obtained during screening confirmed that the OSA patient cohort, on average, reported poor sleep quality. The median Pittsburgh Sleep Quality Index score was 4.5 (3) ($N = 12$), with a score of 5 typically serving as criterion for effectively distinguishing good and poor sleepers (Buysse et al., 1989). Similarly, median Epworth Sleepiness Scale score was 10 (9), which is indicative of higher-normal daytime sleepiness (Johns, 1991).

It is largely held that olfactory stimulation does not perturb sleep and, as discussed earlier, may promote it (Fismer & Pilkington, 2012; Gaeta & Wilson, 2022; Hardy et al., 1995; Kawai et al., 2018; Lytle et al., 2014; Perl et al., 2016; Sano et al., 1998). Nevertheless, to validate that the olfactory stimulation did not induce arousal in our experimental setting, we compared the wake and arousal indices of "Odour" and "Control" nights. We found that "Odour" nights had lower wake index than "Control" nights ("Odour": 1.9 (1.37), "Control": 2.2 (0.7), Wilcoxon signed-rank test (12/17 patients):

TABLE 3 Effects of olfactory stimulation on OSA-related parameters: summary of the results.

Measurement	Odour	Control	z	p		BF_{10}
Session duration (hr)	6.78 (0.85)	6.95 (1.23)	−1.396	0.174	–	0.763
AHI (total)	17.2 (20.9)	28.2 (18.6)	−3.34	0.000846	***	57.9
ODI	12.5 (15.8)	25.7 (25.9)	−3.337	0.000846	***	9.52
Mean SpO_2 (%)	93 (1)	93 (2)	−1.668	0.125	–	0.832
Minimal SpO_2 (%)	84 (4)	82 (5)	1.13	0.271	–	0.451
Time at SpO_2 < 90% (%)	4.4 (4.9)	5.4 (14.0)	−2.51	0.0121	*	3.966
Event duration (s)	20 (4)	20 (4.5)	−1.16	0.263	–	0.26
Wake (index)	1.9 (1.37)	2.2 (0.7)	−2.09	0.036	*	1.958
Arousals (index)	32.9 (26.12)	26.9 (16.05)	−1.585	0.113	–	0.855
Measurement	**Night 1**	**Night 2**	z	p		BF_{10}
AHI (effect of order)	24.7 (22.1)	20.4 (16.8)	0.402	0.687	–	0.32
%TIB at wake (effect of order)	13.2 (8)	8.6 (5.6)	2.627	0.0086	**	6.456
Sleep efficiency (effect of order)	86.8 (8)	91.4 (7.7)	2.627	0.0086	**	5.797

Abbreviations: AHI, apnea–hypopnea index; ODI, oxygen desaturation index; SpO_2, oxygen saturation; %TIB, percentage of time in bed.

$z = -2.0956$, $p = 0.0361$, $BF_{10} = 1.958$; Figure 2d), and arousal scores (both cortical and autonomic) were not significantly different between the two nights (arousal total sleep: "Odour": 32.9 (26.12), "Control": 26.9 (16.05), Wilcoxon signed-rank test (12/17 patients): $z = -1.585$, $p = 0.113$, $BF_{10} = 0.855$). These findings indicate that odour stimulation did not induce arousals and, in line with the literature (Gaeta & Wilson, 2022), reduced the number of awakenings.

Finally, we asked whether olfactory stimulation altered sleep structure, quantified as %TIB spent in wakefulness and in each sleep stage (Table 4). A repeated-measures ANOVA with conditions of "Stimulation" (Odour/Control) and "Stage" (Wake/REM/N1/N2/N3) did not uncover a main effect of "Stimulation" ($F_{1,16} = 0.088$, $p = 0.77$), implying no change in the time spent in each stage between the two nights. In addition, as expected we observed that on the first night in the sleep lab, regardless of whether odours were presented, the %TIB spent at wake was significantly higher (%TIB at wake: "Night 1": 13.2 (8), "Night 2": 8.6 (5.6), Wilcoxon signed-rank test (11/17 patients): $z = 2.6273$, $p = 0.00860$, $BF_{10} = 6.456$) and sleep efficiency was lower (sleep efficiency: "Night 1": 86.8 (8.0), "Night 2": 91.4 (7.7), Wilcoxon signed-rank test (11/17 patients): $z = 2.6273$, $p = 0.00860$, $BF_{10} = 5.797$). Both these factors can be attributed to "first-night" effects due to sleeping in unfamiliar settings (Toussaint et al., 1995). In line with previous studies (Arzi et al., 2010; Badia et al., 1990; Carskadon & Herz, 2004; Perl et al., 2017; Stuck et al., 2007), these findings demonstrate that odour presentation during sleep does not disturb sleep in patients with OSA or alters sleep architecture significantly.

TABLE 4 Sleep structure distribution provided as %TIB, as well as arousals and wake index.

	Odour	Control	p-Value
Wake (%) TIB	13 (9.2)	9.2 (6.2)	0.844
REM (%) TIB	12 (8.5)	13 (7.7)	0.977
N1 (%) TIB	1.8 (4.1)	1.3 (4.5)	0.836
N2 (%) TIB	50.3 (12)	56.1 (9)	0.956
N3 (%) TIB	18.2 (10.6)	16.1 (8.6)	0.402
Arousals	32.9 (23.8)	36.9 (14.9)	0.113
Wake (index)	1.9 (1.3)	2.2 (0.6)	0.036

Note: Values are presented as median (IQR), and the result of a statistical test contrasting between "Odour" and "Control" values.
Abbreviations: REM, rapid eye movement; TIB, time in bed.

5 | DISCUSSION

In this pilot study, we asked whether olfactory stimulation during sleep can alleviate sleep apnea markers. We leveraged on the unique ability of odours to exert physiological and neurological effects during sleep without inducing arousal (Perl et al., 2017; Shanahan & Gottfried, 2014). Our main finding is that transient olfactory stimulation during sleep was associated with a pronounced reduction in patients' AHI. Out of a cohort of 17 patients, 14 patients (82%) exhibited a reduction in AHI while the remaining three (18%) showed very subtle increases (all < 1 AHI). AHI reductions were not related to the severity of baseline OSA markers in this cohort (15 < AHI < 83 at baseline). To our knowledge, this is the first study to report beneficial effects of olfactory stimulation during sleep on OSA in adults.

Our study diverges from previous uses of olfactory stimulation in sleep in two central aspects. First, odours were delivered in transient pulses. Second, we used a rotating selection of odours rather than adhering to one. Both these aspects may have led to less sensory habituation compared with methods of constant passive exposure (Goel et al., 2005; Goel & Lao, 2006). These improvements were achieved thanks to recent technological advancements in conditionally triggered olfactometry (Arzi et al., 2014; Rasch et al., 2007),

miniaturisation (Amores et al., 2019) and on-the-fly processing with machine learning. Compared with the manually validated PSG recording, the olfactometer detected about half of the apnea–hypopnea events detected by PSG. Notably, even though approximately half of the events went undetected, we observed a reduction of over 30% in AHI. We speculate that higher detection rates may result in further reduction in AHI. That said, the dynamics of the respiratory response to odours are unclear and a cumulative effect may be exerted by sparse odour presentations as well.

The severity of OSA is linked to olfactory dysfunction (Salihoglu et al., 2014; Magliulo et al., 2018), yet the exact pathophysiological reasons are unclear (Iannella et al., 2021). During screening, patients reported subjectively normal olfaction but did not undergo an objective evaluation (e.g. "Sniffin" Sticks; Hummel et al., 1997). We speculate the patients with an intact sense of smell will be optimal candidates for this treatment method, given that severe olfactory dysfunction could hinder the perception of incoming sensory stimulation. Follow-up studies should dwell on this point to systematically delineate the precise relationship between olfactory abilities and the influence of olfactory stimulation during sleep on OSA.

Our findings complement evidence from research on neonatal apnea of prematurity (Edraki et al., 2013; Marlier et al., 2005; but see Yaghoubi et al., 2017 for opposite results) who reported ambient exposure to vanillin reduced apnea symptomatology. Here, odour identity did not affect event duration and no single odour type emerged as the most effective. This observation aligns with previous studies showing that during sleep, odour stimulation bears effects regardless of its identity or valence (Perl et al., 2017; Sabiniewicz et al., 2022; Shanahan & Gottfried, 2014). Future studies may systematically test different odorants to tackle this question directly. Olfactory stimulation did not alter sleep structure. However, we cannot disentangle the factors contributing to sleep continuity – the reduction in apneic events and the sleep-promoting effects of odours. We did not observe olfactory-induced improvements in parameters of sleep quality (e.g. %SWS, sleep efficiency).

The neurological mechanisms at play in OSA are unclear. Several characteristics of OSA suggest involvement of the central nervous system in the genesis or maintenance of the syndrome (Morgenthaler et al., 2006). For example, obstruction in OSA is frequently preceded by short breathing pauses, apparently of central origin (Eckert et al., 2009). Which neurological mechanism may mediate the effects of olfactory stimulation on respiration? We propose that the mechanism by which breathing is reinstated is through the aspiration reflex. Nasopharyngeal stimulation was shown to restore spontaneous breathing in rodents and cats (Tomori et al., 1994; Tomori et al., 1995), most likely through medullary nuclei of the pre-Bötzinger complex, a respiratory pacemaker (Batsel & Lines, 1973). Olfactory stimulation may act as a low-amplitude form of nasopharyngeal stimulation, thereby inducing the aspiration reflex without arousal. In line with this proposed mechanism, olfactory stimulation can potentially offer significant relief from central sleep apnea, but this remains to be tested empirically.

Finally, we would like to highlight several limitations of our study. First, due to multiple exclusions, our sample size consisted of 17 patients, which limits the generalizability of the observed effects, and may have obscured additional effects due to low statistical power. Second, our results should be replicated in a more longitudinal design involving multiple nights per condition in order to test gradual cumulative effects and reduce naturally occurring between-night variability in OSA measures. Third, the low number of females in the cohort prevented us from testing any gender-related differences. Lastly, no reports of subjective outcome were collected following the experimental session. We wish to elaborate on the drawback given that subjective outcomes are considered key measurements for treatment efficiency, in tandem with the physiological measurements (Tam et al., 2014). The interaction between subjective and objective measurements is a valuable source of information for evaluation of OSA treatment outcome (Kezirian et al., 2011). We stress the need for the collection of these data in future studies.

In conclusion, the high prevalence of OSA, combined with current insufficient solutions, warrants making the search for alternative treatments a high priority. Our results suggest that contactless transient respiratory-based olfactory stimulation during sleep is a viable such alternative, prompting continued research on this front.

AUTHOR CONTRIBUTIONS

Ofer Perl: Formal analysis; writing - original draft; visualization. **Lilach Kemer:** Investigation; data curation; writing - review and editing. **Amit Green:** Investigation; data curation; writing - review and editing. **Nissim Arish:** Investigation; data curation; writing - review and editing. **Yael Corcos:** Conceptualization; funding acquisition; methodology; writing - review and editing. **Anat Arzi:** Conceptualization; writing – original draft; formal analysis; visualization. **Yaron Dagan:** Conceptualization; investigation; data curation; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

OP and YC are paid consultants at AppScent Medical. LK, AG, NA and YD were responsible for data collection, which was conducted in a blind manner regarding the condition ("odour" or "control") at two clinical centres. The cost of the data collection at the clinical centres was covered by AppScent Medical. OP and AA received the PSG clinical data for analysis only once data collection was terminated, and were not involved in the data collection for the experiment.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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